

SOFTWARE TEST PLAN

BLOOD BANK MANAGEMENT SYSTEM

Technology: Python (Django) / JavaScript (Node.js)

Team Members

PES2UG24CS314 — Nidhi Karkada

PES2UG24CS361 — Pramay Dubey

PES2UG24CS343 — Pavan E Naik

PES2UG24CS326 — Nishit Suresh

1. Introduction

1.1 Purpose

This document is the Software Test Plan for Blood Bank Management System. It outlines the scope, methodology, resource requirements, timeline and criteria for testing of the system based on the functional specifications established by the Software Requirement Specification (SRS) of the project. This serves as a single reference regarding what will be tested, how it will be tested as well as the manner in which test results would be documented and tracked.

1.2 Scope

Testing covers the following functional areas, based on the functional requirements defined in the SRS:

- User Authentication (BBMS-F-001, BBMS-F-002)
- Donor Management (BBMS-F-003 to BBMS-F-006)
- Blood Donation Management (BBMS-F-007 to BBMS-F-009)
- Blood Group Management (BBMS-F-010)
- Blood Inventory Management (BBMS-F-011, BBMS-F-022)
- Blood Availability Search (BBMS-F-012, BBMS-F-013)
- Blood Request Management (BBMS-F-014 to BBMS-F-016)
- Blood Issue and Distribution (BBMS-F-017 to BBMS-F-019)
- Administrative Management (BBMS-F-020, BBMS-F-021)

Along with functional assessment, this plan entails a thorough validation effort (Section 5.1) that relates to authentication, session management, access control and certain input-related vulnerabilities with a reference to SRS security specifications (BBMS-SR-001 to BBMS-SR-005). Since no code sources were available for establishing the implementation status during the plan development, all requirements in the SRS are taken for planned testing scope as opposed to already implemented ones.

1.3 References

- Software Requirements Specification (SRS) — Blood Bank Management System, Version 1.0
- Project Plan — Blood Bank Management System
- IEEE 829 Standard for Software Test Documentation — used as the general structural reference for this test plan, adapted for a college mini-project

1.4 Definitions and Acronyms

- SRS — Software Requirements Specification
- RTM — Requirements Traceability Matrix
- BBMS — Blood Bank Management System
- FR / BBMS-F — Functional Requirement
- SR / BBMS-SR — Security Requirement
- UT / IT / ST — Unit Test / Integration Test / System Test
- SEC — Security test case identifier prefix used in this plan
- RBAC — Role-Based Access Control
- XSS — Cross-Site Scripting
- SQL — Structured Query Language
- DBMS — Database Management System

2. Test Items

The following modules of the Blood Bank Management System (as defined in SRS v1.0) are the items under test:

Module	Description
User Authentication	Login, logout and session handling for all user roles
Donor Management	Donor registration, profile updates, record viewing and eligibility tracking
Blood Donation Management	Recording blood donations and linking them to donor history
Blood Group Management	Maintaining the standard blood group master list used across the system
Blood Inventory Management	Tracking blood unit stock levels and expiry information
Blood Availability Search	Searching current blood availability by blood group
Blood Request Management	Submitting, tracking, approving and rejecting blood requests
Blood Issue and Distribution	Issuing blood against approved requests and updating inventory
Administrative Management	Managing user accounts and filtering/searching system records

3. Features to be Tested

Every feature listed below is mapped to its corresponding SRS functional requirement IDs. Detailed test cases for each are provided in Section 5 (security features) and Appendix A (functional Unit/Integration/System test cases).

Module	Requirement IDs Covered	Number of Test Cases
User Authentication	BBMS-F-001, BBMS-F-002	8
Donor Management	BBMS-F-003, BBMS-F-004, BBMS-F-005, BBMS-F-006	9
Blood Donation Management	BBMS-F-007, BBMS-F-008, BBMS-F-009	8
Blood Group Management	BBMS-F-010	8
Blood Inventory Management	BBMS-F-011, BBMS-F-022	8
Blood Availability Search	BBMS-F-012, BBMS-F-013	8
Blood Request Management	BBMS-F-014, BBMS-F-015, BBMS-F-016	8
Blood Issue and Distribution	BBMS-F-017, BBMS-F-018, BBMS-F-019	8
Administrative Management	BBMS-F-020, BBMS-F-021	8
Security Validation (all modules)	BBMS-SR-001 to BBMS-SR-005	13
Total	—	86

4. Features Not to be Tested

The following are outside the scope of this test plan, either because they are outside the SRS scope or because they require infrastructure/tools beyond a college mini-project:

- Hospital patient management, medical diagnosis, or clinical decision support — explicitly outside the SRS scope.
- External courier/logistics tracking for blood transport — explicitly outside the SRS scope.
- Load/performance testing at production scale (e.g. thousands of concurrent users) — not applicable to a college-level deployment.
- Penetration testing using specialized security tools — security validation in this plan is limited to manual functional checks (Section 5.1), not a full security audit.

- HTTPS/TLS certificate configuration testing — treated as a deployment/infrastructure concern rather than an application test case (see BBMS-SR-003 in the RTM).
- Third-party integrations not defined in the SRS (e.g. payment gateways, SMS/email notification services) — no such integrations are specified.
- Mobile application testing — the SRS defines a web-based application only.

Assumption: any feature not explicitly listed in the SRS functional requirements is assumed to be out of scope for this project and this test plan.

5. Test Approach / Test Strategy

This section describes the overall approach used to test the system, including the levels of testing applied, the manual testing approach, and the criteria used to begin and end test execution. Dedicated security validation is described separately in Section 5.1.

Testing Levels

Three levels of testing are applied to this project: Unit Testing, Integration Testing and System Testing.

Unit Testing

Unit Testing (UT) verifies individual validation rules and logic within a single module, such as form field validation, independent of other modules.

Integration Testing

Integration Testing (IT) verifies that connected components work together correctly, such as a form submission being saved to the database or a donation updating inventory.

System Testing

System Testing (ST) verifies complete end-to-end workflows from the user's point of view, across multiple modules.

Manual Testing Approach

The test is meant to be performed manually through the user interface of the application and involves using the test data mentioned in the respective test case. Each test case is performed step by step, and the result is compared with the expected outcome in order to determine the pass or fail status of the test case.

Entry Criteria

- The SRS and Project Plan have been reviewed and approved by the team.
- Test cases have been designed and reviewed for the module(s) under test.
- The module(s) under test have been implemented and deployed to the test environment.
- Required test data has been prepared.
- The test environment (application, database, browser) is available and accessible.

Exit Criteria

- All planned test cases for the module(s) under test have been executed.
- No critical or high-severity defects remain open for the module(s) under test.
- Requirement traceability (Section 13) shows every in-scope requirement mapped to at least one executed test case.
- Test results have been recorded and the test summary has been reviewed by the team.

5.1 Security Validation

The emphasis of this part is on providing a comprehensive explanation of the security validation process adopted for the Blood Bank Management System. This process includes several essential aspects such as authentication and login security, password encryption, session handling, authorization and role-based access control, input validation practices, and methods of preventing SQL injection and Cross Site Scripting (XSS). The validation action during the security testing process has been well-defined and organized by creating a traceability mapping between the security test cases and different security requirements specified in the software requirements specification (BBMS-SR-001 to BBMS-SR-005) document along with other functional requirements.

- Authentication and login security
- Password protection and masking
- Session management
- Authorization and role-based access control
- Input validation
- SQL Injection prevention
- Cross-Site Scripting (XSS) prevention
- Unauthorized data access prevention
- Audit/logging verification

Note: Session-timeout behavior (SEC-06) and the implementation of HTTPS protocol (BBMS-SR-003) have been included here for purposes of keeping the paper up-to-date in terms of security practices; however, the proposed SRS shall not set the specific time limit on the session timeout. What is more, the HTTPS implementation is approached as a control related to deployment/configuration and not an application testing case (see Section 13, RTM).

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
SEC-01	Security - Authentication	Verify an unauthenticated user cannot access a protected page by entering its URL directly	User is not logged in	1. Without logging in, enter the URL of a protected page (e.g. donor dashboard) directly into the browser.	N/A	User should be redirected to the login page and denied access to the protected page	Not executed	Not Run
SEC-02	Security - Authentication	Verify login failure messages do not reveal whether the username or password was incorrect	Login page is loaded	1. Enter an unregistered username with any password. 2. Note the error message. 3. Enter a registered username with a wrong password. 4. Compare error messages.	Unregistered: nouser@test.com Registered: donor1@test.com / Wrong@123	Both cases should show the same generic error message, without indicating which field was incorrect	Not executed	Not Run
SEC-03	Security - Password Protection	Verify the password field masks characters as they are typed on the login and registration forms	Login/registration page is loaded	1. Click into the password field. 2. Type a password.	Password: Test@123	Characters should appear masked (e.g. as dots/asterisks), not as plain text	Not executed	Not Run
SEC-04	Security - Password Protection	Verify passwords are stored using a hashing mechanism rather than in plain text	A user account exists	1. Inspect the stored user record in the database (e.g. via the database admin tool).	N/A	The stored password value should be a hashed string, not the original plain-text password	Not executed	Not Run
SEC-05	Security - Session Management	Verify a user's session ends on logout and cannot be restored using the browser back button	User is logged in	1. Click Logout. 2. Click the browser back button. 3. Attempt to interact with the page.	N/A	User should not regain access to the protected page; the system should require login again	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
SEC-06	Security - Session Management	Verify a session does not remain indefinitely active after a period of user inactivity	User is logged in	1. Log in. 2. Leave the session idle for the configured timeout period (if implemented). 3. Attempt to perform an action.	N/A	An appropriate session-timeout / re-authentication prompt should be shown if session timeout is implemented	Not executed	Not Run
SEC-07	Security - Authorization / RBAC	Verify a Donor or Requester account cannot access Administrator pages by entering the URL directly	A Donor/Requester account is logged in	1. Log in as a Donor or Requester. 2. Enter the URL of an administrator-only page directly.	N/A	Access should be denied and the user redirected away from the administrative page	Not executed	Not Run
SEC-08	Security - Authorization / RBAC	Verify only Administrator/Staff roles can approve or reject a blood request	A pending blood request exists; a non-staff account is logged in	1. Log in as a Donor or Requester. 2. Attempt to approve/reject a pending request directly (e.g. via request manipulation).	N/A	The action should be rejected; only Administrator/Staff accounts should be able to approve or reject requests	Not executed	Not Run
SEC-09	Security - Input Validation	Verify text input fields reject or safely handle script/special-character injection attempts	A form with a text field (e.g. donor name) is open	1. Enter a value containing script tags or special characters into a text field. 2. Submit the form.	Name: <script>alert(1)</script>	The input should be rejected, sanitized, or safely escaped; it should not be executed or stored as active script	Not executed	Not Run
SEC-10	Security - SQL Injection Prevention	Verify the login form rejects common SQL injection patterns without granting access	Login page is loaded	1. Enter a SQL injection pattern in the username or password field. 2. Click Login.	Username: admin' OR '1'=1 Password: anything	Login should be rejected; the injection pattern should not bypass authentication or expose database errors	Not executed	Not Run
SEC-11	Security - XSS Prevention	Verify a submitted text value containing a script tag is not executed when displayed back to a user	A form allowing free text (e.g. request reason) is available	1. Submit a value containing a script tag as the input. 2. View the page where this value is later displayed (e.g. request details).	Reason: <script>alert('xss')</script>	The script should not execute; the value should be displayed as plain text or safely escaped	Not executed	Not Run
SEC-12	Security - Unauthorized Data Access Prevention	Verify a logged-in donor cannot view another donor's profile by modifying an ID parameter in the URL	Two donor accounts exist; one is logged in	1. Log in as Donor A. 2. Open Donor A's profile page and note the URL/ID. 3. Modify the ID in the URL to reference Donor B's record.	Donor A ID vs Donor B ID	Access should be denied, or the system should not display Donor B's private information to Donor A	Not executed	Not Run
SEC-13	Security - Audit / Logging	Verify a change to inventory or request status generates an audit log entry with user, action and timestamp	Staff is logged in; an inventory or request record exists	1. Perform an action that changes inventory or request status (e.g. approve a request). 2. Open the audit/activity log.	N/A	A new audit log entry should appear recording the user, the action performed, and the date/time	Not executed	Not Run

6. Test Environment

Testing is intended to be carried out in the following environment:

- Windows or Linux development environment
- Standard web browser (e.g. Chrome or Firefox) for manual UI testing
- Python / Django backend
- JavaScript / Node.js for supporting backend services
- Relational database used by the project (e.g. MySQL)
- Application running on a local development server (localhost)

7. Test Schedule

The table below shows a realistic four-week testing timeline within the overall project schedule, covering test design through final reporting. 'X' marks the weeks during which an activity is active.

Testing Activity	Week 1	Week 2	Week 3	Week 4
Test planning & test case design	X			
Unit testing	X	X		
Integration testing		X	X	
System testing			X	
Security validation			X	
Regression / bug-fix retesting			X	X
Test reporting & sign-off				X

8. Test Deliverables

- This Software Test Plan document
- Test case specifications (Unit, Integration, System and Security – Section 5.1 and Appendix A)
- Test data sets used for execution
- Test execution summary and metrics report (Section 14)
- Defect log (once testing begins)
- Requirements Traceability Matrix (Section 13)
- Final test sign-off / approval record (Section 15)

9. Roles and Responsibilities

Tasks are allocated among test planning, test design, test data preparation, manual execution, defect logging, and documentation. The table below shows the roles in the IEEE style. The actual division of work based on each testing output is presented below.

9.1 Roles Overview

Role	Team Member	Primary Responsibility
Test Lead / Planning	Nidhi Karkada (PES2UG24CS314)	Test plan structure, scope, entry/exit criteria, authentication & donor test design and execution
Test Analyst – Design	Pramay Dubey (PES2UG24CS361)	Integration test design, test data preparation, donation/blood group modules, documentation
Test Engineer – Execution	Pavan E Naik (PES2UG24CS343)	Unit/integration test design for inventory & search, defect/result recording
Test Reporting Lead	Nishit Suresh (PES2UG24CS326)	System test design for request/issue/admin modules, traceability, final test report

9.2 Overall Division of Work

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Test planning, unit test design for Authentication and Donor Management, manual execution of authentication/donor test cases
Pramay Dubey (PES2UG24CS361)	Integration test design for Donation and Blood Group modules, test data preparation, user-facing documentation
Pavan E Naik (PES2UG24CS343)	Unit and integration test design for Inventory and Availability Search, defect/result recording
Nishit Suresh (PES2UG24CS326)	System test design for Request, Issue/Distribution and Administrative workflows, final test report preparation

9.3 Division of Work by Testing Deliverable

Test Planning

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Test plan overview and testing scope
Pramay Dubey (PES2UG24CS361)	Testing types and test environment description
Pavan E Naik (PES2UG24CS343)	Test case format and organization
Nishit Suresh (PES2UG24CS326)	Requirement traceability structure

Unit Test Cases

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Authentication and Donor Management unit tests
Pramay Dubey (PES2UG24CS361)	Blood Donation and Blood Group Management unit tests
Pavan E Naik (PES2UG24CS343)	Blood Inventory and Availability Search unit tests
Nishit Suresh (PES2UG24CS326)	Blood Request, Issue/Distribution and Administrative unit tests

Integration Test Cases

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Authentication and Donor Management integration tests
Pramay Dubey (PES2UG24CS361)	Blood Donation and Blood Group Management integration tests
Pavan E Naik (PES2UG24CS343)	Blood Inventory and Availability Search integration tests
Nishit Suresh (PES2UG24CS326)	Blood Request, Issue/Distribution and Administrative integration tests

System Test Cases

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Authentication and Donor Management system tests
Pramay Dubey (PES2UG24CS361)	Blood Donation and Blood Group Management system tests
Pavan E Naik (PES2UG24CS343)	Blood Inventory and Availability Search system tests
Nishit Suresh (PES2UG24CS326)	Blood Request, Issue/Distribution and Administrative system tests

Test Data Preparation

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Sample donor and user account data
Pramay Dubey (PES2UG24CS361)	Sample donation and blood group data
Pavan E Naik (PES2UG24CS343)	Sample inventory and expiry data
Nishit Suresh (PES2UG24CS326)	Sample request and issue transaction data

Manual Test Execution

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Execute authentication and donor management test cases
Pramay Dubey (PES2UG24CS361)	Execute donation and blood group management test cases
Pavan E Naik (PES2UG24CS343)	Execute inventory and availability search test cases
Nishit Suresh (PES2UG24CS326)	Execute request, issue/distribution and administrative test cases

Defect / Result Recording

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Record results for authentication and donor management tests
Pramay Dubey (PES2UG24CS361)	Record results for donation and blood group management tests
Pavan E Naik (PES2UG24CS343)	Record results for inventory and availability search tests
Nishit Suresh (PES2UG24CS326)	Record results for request, issue/distribution and administrative tests, and consolidate defect list

Final Test Documentation

Team Member	Assigned Responsibility
Nidhi Karkada (PES2UG24CS314)	Compile test execution summary
Pramay Dubey (PES2UG24CS361)	Review test data and expected results for consistency
Pavan E Naik (PES2UG24CS343)	Review requirement traceability table
Nishit Suresh (PES2UG24CS326)	Final formatting and consolidation of the test plan document

10. Risks and Mitigation

Risk	Impact	Mitigation
Limited available time due to parallel coursework/internship commitments	Medium	Follow the WBS/schedule and prioritize core modules first
Modules not implemented/available at the time testing is due to begin	High	Test cases are pre-designed against the SRS so execution can start as soon as each module is ready
Team member unavailability during a testing phase	Medium	Responsibilities and module knowledge are documented (Section 9) so another member can step in
Test environment / configuration mismatches between members	Medium	Use a standardized environment setup and shared configuration (Section 6)
Ambiguous or changing requirements	Low	Test cases are reviewed against the SRS and tracked via the RTM (Section 13) to catch gaps early
Security test cases require unfamiliar concepts (e.g. SQL injection, XSS)	Medium	Team review of standard security testing references before execution of Section 5.1 cases

11. Assumptions and Dependencies

11.1 Assumptions

- The implementation will follow the SRS functional and security requirements as written.
- The test environment mirrors the development environment described in Section 6.
- Test data used is synthetic and does not contain real personal information.
- All four team members have access to a shared repository and test environment.

11.2 Dependencies

- A working build of each module must be available before its corresponding test cases can be executed.
- The database schema must match the design described in the Project Plan.
- Team members must complete their assigned development modules according to the Project Plan schedule before those modules can be tested.

12. Suspension and Resumption Criteria

12.1 Suspension Criteria

- Testing of a module will be suspended if a blocking defect prevents further execution (e.g. application crash, module fails to load).
- Testing will be suspended if the test environment becomes unavailable (e.g. server or database down).

12.2 Resumption Criteria

- Testing will resume once the blocking defect has been fixed and verified, or the environment has been restored.
- Test cases affected by the fix will be re-executed (regression) before continuing with remaining new test cases.

13. Requirements Traceability Matrix (RTM)

The table below maps every SRS requirement (functional and security) to the test case IDs that cover it and the corresponding test level(s).

SRS Requirement ID	Requirement Description	Test Case IDs	Test Level
BBMS-F-001	User login	UT-AUTH-01, UT-AUTH-02, UT-AUTH-03, IT-AUTH-01, IT-AUTH-02, IT-AUTH-03, ST-AUTH-01, SEC-01, SEC-02, SEC-10	Unit, Integration, System, Security
BBMS-F-002	User logout / session end	ST-AUTH-02, SEC-05	System, Security
BBMS-F-003	Donor registration	UT-DONOR-01, UT-DONOR-02, UT-DONOR-03, UT-DONOR-04, IT-DONOR-01, IT-DONOR-02, ST-DONOR-01, SEC-09, SEC-11	Unit, Integration, System, Security
BBMS-F-004	Update donor information	IT-DONOR-03	Integration
BBMS-F-005	View donor records	ST-DONOR-01, SEC-12	System, Security
BBMS-F-006	Mark donor eligibility status	IT-DONOR-04	Integration
BBMS-F-007	Record blood donation	UT-DNTN-01, UT-DNTN-02, UT-DNTN-03, IT-DNTN-01, IT-DNTN-03, ST-DNTN-01	Unit, Integration, System
BBMS-F-008	View donation history	ST-DNTN-02	System
BBMS-F-009	Auto-update inventory after donation	IT-DNTN-02	Integration
BBMS-F-010	Maintain blood group master list	UT-BG-01, UT-BG-02, UT-BG-03, UT-BG-04, IT-BG-01, IT-BG-02, ST-BG-01, ST-BG-02	Unit, Integration, System
BBMS-F-011	View inventory summary	UT-INV-01, IT-INV-01, IT-INV-03, ST-INV-01	Unit, Integration, System
BBMS-F-022	Record expiry date / exclude expired units	UT-INV-02, IT-INV-02, ST-INV-02	Unit, Integration, System
BBMS-F-012	Search blood availability by group	UT-SRCH-01, UT-SRCH-03, IT-SRCH-01, ST-SRCH-01	Unit, Integration, System
BBMS-F-013	View available blood units	UT-SRCH-02, IT-SRCH-02, IT-SRCH-03, ST-SRCH-02	Unit, Integration, System
BBMS-F-014	Submit blood request	UT-REQ-01, UT-REQ-02, UT-REQ-03, IT-REQ-01, ST-REQ-01	Unit, Integration, System
BBMS-F-015	View / track request status	IT-REQ-02	Integration
BBMS-F-016	Approve or reject blood request	IT-REQ-03, ST-REQ-02, SEC-08	Integration, System, Security
BBMS-F-017	Issue blood against approved request	UT-ISS-01, UT-ISS-02, IT-ISS-01, ST-ISS-01	Unit, Integration, System
BBMS-F-018	Auto-reduce inventory after issue	IT-ISS-02	Integration
BBMS-F-019	Maintain issue / distribution records	UT-ISS-03, IT-ISS-03, ST-ISS-02, SEC-13	Unit, Integration, System, Security
BBMS-F-020	Manage user accounts (create / deactivate)	UT-ADM-01, UT-ADM-02, IT-ADM-01, IT-ADM-02, ST-ADM-01, ST-ADM-02, SEC-07	Unit, Integration, System, Security
BBMS-F-021	Filter / search records	UT-ADM-03, IT-ADM-03	Unit, Integration
BBMS-SR-001	Require valid login credentials before access	SEC-01, SEC-02, SEC-03, SEC-05, SEC-06, SEC-10	Security
BBMS-SR-002	Restrict admin functions to Administrator / Staff role	SEC-07, SEC-08, SEC-12	Security
BBMS-SR-003	Use HTTPS for all communication	Not covered by a dedicated manual test case (deployment/configuration-level control)	N/A
BBMS-SR-004	Store passwords using secure hashing	SEC-04	Security
BBMS-SR-005	Record audit entry for inventory / request changes	SEC-13	Security

14. Test Metrics and Reporting

This part of the document outlines the various metrics that will be monitored after test execution is initiated. As none of the test cases have been executed at this moment, the current state of metrics indicates that the test execution has not started yet, instead of showing any manipulative results.

14.1 Requirement Coverage

Requirement coverage = (requirements with at least one mapped test case / total requirements) × 100. Based on the RTM in Section 13: functional requirement coverage is 22/22 (100%); security requirement coverage is 4/5 (80%), as BBMS-SR-003 (HTTPS) is a deployment-level control rather than a dedicated test case.

14.2 Test Execution Percentage

Test execution percentage = (executed test cases / total test cases) × 100. Current status: 0 of 86 test cases executed (0%), as testing has not yet commenced.

14.3 Pass / Fail Percentage

Pass/fail percentage will be calculated as (passed / executed) × 100 and (failed / executed) × 100 once execution begins. Not applicable while 0 test cases have been executed.

14.4 Defect Density

Defect density (defects found per module, or per number of test cases in a module) will be calculated once testing begins and defects are logged. No defects have been recorded yet.

14.5 Current Execution Status

Test Type	Total	Passed	Failed	Not Run
Unit Testing	29	0	0	29
Integration Testing	27	0	0	27
System Testing	17	0	0	17
Security Testing	13	0	0	13
Total	86	0	0	86

14.6 Defect Status Tracking

No defects have been logged yet, as test execution has not commenced. The table structure below will be used to track defects once testing begins.

Defect ID	Description	Module	Severity	Status
—	No defects recorded yet	—	—	—

15. Approvals

Role	Name	Signature / Email	Date
Course Coordinator	—	—	—

Appendix A: Detailed Test Case Specifications

This appendix preserves the complete set of Unit, Integration and System test cases originally prepared for this project, unchanged. Security test cases are provided in Section 5.1.

A.1 Unit Test Cases

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT-AUTH-01	User Authentication	Validate login form rejects empty username field	Login page is loaded	1. Leave username field blank. 2. Enter password. 3. Click Login.	Username: (blank) Password: Test@123	An appropriate validation error should be displayed and login should not proceed	Not executed	Not Run
UT-AUTH-02	User Authentication	Validate login form rejects empty password field	Login page is loaded	1. Enter username. 2. Leave password blank. 3. Click Login.	Username: donor1@test.com Password: (blank)	An appropriate validation error should be displayed and login should not proceed	Not executed	Not Run
UT-AUTH-03	User Authentication	Validate password field masks entered characters	Login page is loaded	1. Click the password field. 2. Type a password.	Password: Test@123	Entered characters should be masked and not shown in plain text	Not executed	Not Run
UT-DONDR-01	Donor Management	Validate registration form rejects missing name field	Registration page is loaded	1. Leave Name blank. 2. Fill remaining fields with valid data. 3. Click Register.	Name: (blank) Email: donor2@test.com Phone: 9876543210 Blood Group: O+	An appropriate validation error should be displayed and registration should not proceed	Not executed	Not Run
UT-DONDR-02	Donor Management	Validate registration form rejects invalid email format	Registration page is loaded	1. Enter Name. 2. Enter an invalid email format. 3. Click Register.	Email: donor2test.com	An appropriate validation error should be displayed for invalid email format	Not executed	Not Run
UT-DONDR-03	Donor Management	Validate registration form rejects invalid phone number format	Registration page is loaded	1. Enter Name and Email. 2. Enter a phone number with letters or wrong length. 3. Click Register.	Phone: 98A8543210	An appropriate validation error should be displayed for invalid phone number	Not executed	Not Run
UT-DONDR-04	Donor Management	Validate registration form rejects missing blood group selection	Registration page is loaded	1. Fill all fields except Blood Group. 2. Click Register.	Blood Group: (not selected)	An appropriate validation error should be displayed requiring blood group selection	Not executed	Not Run
UT-DNTN-01	Blood Donation Management	Validate donation form rejects missing donor selection	Staff is on the donation recording page	1. Leave donor field unselected. 2. Enter quantity and date. 3. Click Save.	Donor: (not selected) Quantity: 1 unit	An appropriate validation error should be displayed requiring donor selection	Not executed	Not Run
UT-DNTN-02	Blood Donation Management	Validate donation form rejects zero or negative quantity	Staff is on the donation recording page	1. Select a donor. 2. Enter quantity as 0 or negative. 3. Click Save.	Quantity: 0	An appropriate validation error should be displayed for invalid quantity	Not executed	Not Run
UT-DNTN-03	Blood Donation Management	Validate donation date cannot be a future date	Staff is on the donation recording page	1. Select a donor. 2. Enter a future date. 3. Click Save.	Donation Date: a future date	An appropriate validation error should be displayed for an invalid donation date	Not executed	Not Run
UT-BG-01	Blood Group Management	Validate only standard blood groups are selectable in forms	A form with a blood group field is open	1. Open the blood group dropdown. 2. Review the listed options.	N/A	Dropdown should list exactly the eight standard blood groups (A+, A-, B+, B-, AB+, AB-, O+, O-)	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT-BG-02	Blood Group Management	Validate blood group field rejects a value outside the standard list	A form allows blood group entry	1. Attempt to submit a value outside the standard list.	Blood Group: X+	An appropriate validation error should be displayed for an invalid blood group value	Not executed	Not Run
UT-BG-03	Blood Group Management	Validate blood group list contains no duplicate entries	Blood group dropdown is open	1. Open blood group dropdown. 2. Check for duplicate entries.	N/A	Each of the eight blood groups should appear exactly once	Not executed	Not Run
UT-BG-04	Blood Group Management	Validate blood group values are normalized (e.g. trimmed of extra spaces)	A blood group value is submitted with extra spaces	1. Submit a blood group value with leading/trailing spaces.	Blood Group: " O+ "	Value should be normalized/trimmed and matched correctly to "O+"	Not executed	Not Run
UT-INV-01	Blood Inventory Management	Validate inventory summary shows zero stock rather than omitting a blood group	No units recorded for a specific blood group	1. Open inventory summary page. 2. Check the entry for the blood group with no stock.	Blood Group: AB- (assume 0 units)	Inventory summary should display 0 units for that blood group rather than omitting it	Not executed	Not Run
UT-INV-02	Blood Inventory Management	Validate expiry date cannot be earlier than the unit's collection date	Staff is recording expiry information for a unit	1. Enter an expiry date earlier than the unit's collection date. 2. Click Save.	Expiry Date: earlier than collection date	An appropriate validation error should be displayed for an invalid expiry date	Not executed	Not Run
UT-INV-03	Blood Inventory Management	Validate inventory count cannot be manually set to a negative value	Staff attempts to directly edit inventory count	1. Attempt to set inventory count to a negative number. 2. Click Save.	Units: -5	An appropriate validation error should be displayed and negative values should be rejected	Not executed	Not Run
UT-SRCH-01	Blood Availability Search	Validate search requires a blood group to be selected	Search page is open	1. Leave blood group unselected. 2. Click Search.	Blood Group: (not selected)	An appropriate validation error should be displayed requiring blood group selection	Not executed	Not Run
UT-SRCH-02	Blood Availability Search	Validate search shows a zero-result message for a blood group with no stock	A blood group has zero available units	1. Select a blood group with zero stock. 2. Click Search.	Blood Group: AB- (0 units)	Search should indicate that no units are currently available for the selected blood group	Not executed	Not Run
UT-SRCH-03	Blood Availability Search	Validate search handles an unsupported blood group value gracefully	Search page is open	1. Attempt to submit a search with an unsupported value.	Blood Group: "X+"	An appropriate validation error should be displayed rather than a system error	Not executed	Not Run
UT-REQ-01	Blood Request Management	Validate request form rejects missing blood group	Request page is open	1. Leave blood group unselected. 2. Enter quantity and reason. 3. Click Submit.	Blood Group: (not selected)	An appropriate validation error should be displayed requiring blood group selection	Not executed	Not Run
UT-REQ-02	Blood Request Management	Validate request form rejects zero or negative quantity	Request page is open	1. Select blood group. 2. Enter quantity as 0. 3. Click Submit.	Quantity: 0	An appropriate validation error should be displayed for invalid quantity	Not executed	Not Run
UT-REQ-03	Blood Request Management	Validate request form rejects missing reason field	Request page is open	1. Fill blood group and quantity. 2. Leave reason blank. 3. Click Submit.	Reason: (blank)	An appropriate validation error should be displayed if reason is a required field	Not executed	Not Run
UT-ISS-01	Blood Issue and Distribution	Validate blood cannot be issued against a request that is not approved	A pending (not approved) request exists	1. Attempt to issue blood against the pending request.	Request Status: Pending	An appropriate validation error should be displayed preventing issue against a non-approved request	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT-ISS-02	Blood Issue and Distribution	Validate issued quantity cannot exceed the approved request quantity	An approved request for a specific quantity exists	1. Attempt to issue a quantity greater than approved. 2. Click Issue.	Approved: 2 units Attempted Issue: 5 units	An appropriate validation error should be displayed preventing over-issue	Not executed	Not Run
UT-ISS-03	Blood Issue and Distribution	Validate issue date cannot be earlier than the request approval date	Staff is recording an issue transaction	1. Attempt to enter an issue date earlier than the approval date.	Issue Date: earlier than approval date	An appropriate validation error should be displayed for an invalid issue date	Not executed	Not Run
UT-ADM-01	Administrative Management	Validate new user account form rejects missing required fields	Administrator is on the user account creation page	1. Leave a required field (e.g. username) blank. 2. Click Create.	Username: (blank)	An appropriate validation error should be displayed and the account should not be created	Not executed	Not Run
UT-ADM-02	Administrative Management	Validate duplicate username is rejected when creating a user account	A user account with the given username already exists	1. Enter a username that already exists. 2. Click Create.	Username: staff01 (already exists)	An appropriate validation error should be displayed indicating the username already exists	Not executed	Not Run
UT-ADM-03	Administrative Management	Validate filter/search handles no matching records gracefully	Search page is open	1. Enter a search term that matches no records. 2. Click Search.	Search Term: "ZZZZZ"	An appropriate "no records found" message should be displayed	Not executed	Not Run

A.2 Integration Test Cases

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT-AUTH-01	User Authentication	Verify submitted credentials are checked against stored user records	A registered user account exists	1. Enter registered username. 2. Enter matching password. 3. Click Login.	Username: donor1@test.com Password: Test@123	Backend should authenticate the user and grant access to the corresponding dashboard	Not executed	Not Run
IT-AUTH-02	User Authentication	Verify an incorrect password is rejected by the backend	A registered user account exists	1. Enter registered username. 2. Enter an incorrect password. 3. Click Login.	Username: donor1@test.com Password: Wrong@123	Login should be rejected with an appropriate error message	Not executed	Not Run
IT-AUTH-03	User Authentication	Verify role-based dashboard redirection after login	Donor and Administrator accounts exist	1. Log in with a Donor account, note redirected page. 2. Log out. 3. Log in with an Administrator account, note redirected page.	Donor: donor1@test.com/Test@123 Admin: admin1@test.com/Admin@123	Donor should land on donor dashboard; Administrator should land on administrator dashboard	Not executed	Not Run
IT-DONOR-01	Donor Management	Verify valid donor registration is saved to the donor database	Registration page is loaded	1. Enter valid donor details. 2. Click Register. 3. Check donor list.	Name: Test Donor Email: donor2@test.com Phone: 9876543210 Blood Group: O+	Donor record should be created and visible in the donor list	Not executed	Not Run
IT-DONOR-02	Donor Management	Verify duplicate donor registration (same email) is handled	A donor with the given email is already registered	1. Enter donor details using an already-registered email. 2. Click Register.	Email: donor2@test.com (already registered)	An appropriate validation error should be displayed indicating the donor/email already exists	Not executed	Not Run
IT-DONOR-03	Donor Management	Verify donor profile update is saved correctly	Donor is registered and logged in	1. Open donor profile. 2. Update phone number. 3. Save changes. 4. Reopen profile.	Updated Phone: 9123456780	Updated details should be saved and displayed correctly on reload	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT-DONOR-04	Donor Management	Verify staff can mark a donor's eligibility status	Staff is logged in; donor record exists	1. Open donor record. 2. Set eligibility status. 3. Save.	Eligibility: Temporarily Ineligible	Donor eligibility status should be updated and visible on the donor record	Not executed	Not Run
IT-DNTN-01	Blood Donation Management	Verify a recorded donation is saved and linked to the correct donor	A registered donor exists	1. Select the donor. 2. Enter blood group and quantity. 3. Click Save. 4. Open donor's donation history.	Blood Group: B+ Quantity: 1 unit	Donation entry should be saved and visible in the donor's donation history	Not executed	Not Run
IT-DNTN-02	Blood Donation Management	Verify recording a donation automatically updates blood inventory	Current inventory count for a blood group is known	1. Note current inventory for a blood group. 2. Record a new donation for that group. 3. Check inventory again.	Blood Group: B+ Quantity: 1 unit	Inventory count for the blood group should increase by the donated quantity	Not executed	Not Run
IT-DNTN-03	Blood Donation Management	Verify donation recording is rejected for a non-existent donor reference	Staff is on donation recording page	1. Attempt to submit a donation referencing a donor ID that does not exist.	Donor ID: invalid/non-existent	System should reject the request with an appropriate error rather than saving invalid data	Not executed	Not Run
IT-BG-01	Blood Group Management	Verify the blood group list is consistent across donation, inventory and request forms	Blood group master list is defined	1. Open donation form blood group list. 2. Open inventory blood group list. 3. Open request form blood group list. 4. Compare options.	N/A	The same eight blood groups should appear consistently across all forms	Not executed	Not Run
IT-BG-02	Blood Group Management	Verify blood group list is unaffected by repeated use in donations/requests	Several donations/requests have been recorded	1. Record several donations and requests using different blood groups. 2. Reopen the blood group dropdown.	N/A	The standard eight-value blood group list should remain unchanged	Not executed	Not Run
IT-INV-01	Blood Inventory Management	Verify inventory reflects combined effect of donations and issues	A blood group has had both a donation and an issue recorded	1. Record a donation for a blood group. 2. Record an issue for the same blood group. 3. Check inventory summary.	Blood Group: O+ Donated: 2 units Issued: 1 unit	Inventory should reflect net stock (donations minus issues) accurately	Not executed	Not Run
IT-INV-02	Blood Inventory Management	Verify a unit past its expiry date is excluded from available inventory	A blood unit has an expiry date in the past	1. Record a unit with an expiry date in the past. 2. Open the availability/inventory view.	Expiry Date: a past date	The expired unit should not be counted in available inventory	Not executed	Not Run
IT-INV-03	Blood Inventory Management	Verify inventory is unchanged after a rejected blood request	A blood request is submitted and then rejected	1. Submit a blood request. 2. Reject the request. 3. Check inventory.	N/A	Inventory should remain unchanged since no units were issued	Not executed	Not Run
IT-SRCH-01	Blood Availability Search	Verify search results match current inventory data	Inventory has a known unit count for a blood group	1. Note current inventory for a blood group. 2. Search availability for that group. 3. Compare results.	Blood Group: O+	Search result should match the current inventory count exactly	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT-SRCH-02	Blood Availability Search	Verify search results update after a new donation is recorded	Inventory count for a blood group is known	1. Search availability for a blood group. 2. Record a new donation for that group. 3. Search again.	Blood Group: B+	Search result after the donation should reflect the increased unit count	Not executed	Not Run
IT-SRCH-03	Blood Availability Search	Verify search results exclude expired blood units	An expired unit exists for a blood group	1. Search availability for the blood group with an expired unit.	N/A	Expired units should not be included in the displayed available count	Not executed	Not Run
IT-REQ-01	Blood Request Management	Verify a submitted request is saved with status "Pending"	Requester is logged in	1. Submit a valid blood request. 2. Check request status.	Blood Group: O- Quantity: 2 units	Request should be saved and initially show status "Pending"	Not executed	Not Run
IT-REQ-02	Blood Request Management	Verify requester can view the current status of their submitted request	A request has been submitted	1. Log in as the requester. 2. Open "My Requests" page.	N/A	Requester should see the request with its current status (Pending/Approved/Rejected/Issued)	Not executed	Not Run
IT-REQ-03	Blood Request Management	Verify staff approval of a request updates its status correctly	A pending request exists	1. Log in as staff. 2. Open the pending request. 3. Click Approve. 4. Check request status.	N/A	Request status should change to "Approved" and be visible to the requester	Not executed	Not Run
IT-ISS-01	Blood Issue and Distribution	Verify issuing blood against an approved request marks it as "Issued"	An approved request exists	1. Staff opens the approved request. 2. Clicks Issue. 3. Checks request status.	N/A	Request status should change to "Issued"	Not executed	Not Run
IT-ISS-02	Blood Issue and Distribution	Verify issuing blood automatically reduces inventory for the corresponding blood group	Inventory count is known; an approved request exists	1. Note current inventory. 2. Issue blood against the approved request. 3. Check inventory again.	Blood Group: O+ Issued: 1 unit	Inventory count should decrease by the issued quantity	Not executed	Not Run
IT-ISS-03	Blood Issue and Distribution	Verify an issue transaction is recorded in the issue/distribution log	An issue has just been performed	1. Perform a blood issue. 2. Open the issue/distribution records.	N/A	A new entry should appear with date, quantity, blood group and requester details	Not executed	Not Run
IT-ADM-01	Administrative Management	Verify administrator can deactivate an existing user account	An active user account exists	1. Administrator opens the account. 2. Clicks Deactivate. 3. Checks account status.	N/A	Account status should change to inactive/deactivated	Not executed	Not Run
IT-ADM-02	Administrative Management	Verify a deactivated user cannot log in	A user account has been deactivated	1. Attempt to log in using the deactivated account's credentials.	Username: staff01 (deactivated)	Login should be rejected with an appropriate message	Not executed	Not Run
IT-ADM-03	Administrative Management	Verify filter/search on donor records returns only matching results	Multiple donor records exist	1. Enter a search term (e.g. blood group or name). 2. Click Search/Filter.	Search Term: "O+"	Only donor records matching the search term should be displayed	Not executed	Not Run

A.3 System Test Cases

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST-AUTH-01	User Authentication	End-to-end valid login workflow	Login page is loaded; a valid account exists	1. Open the login page. 2. Enter valid username and password. 3. Click Login.	Username: donor1@test.com Password: Test@123	User should be logged in and the dashboard should load correctly	Not executed	Not Run
ST-AUTH-02	User Authentication	End-to-end logout workflow	User is logged in	1. Click Logout. 2. Use the browser back button to try to reach a protected page.	N/A	User should be logged out; protected pages should not be accessible without logging in again	Not executed	Not Run
ST-DONDR-01	Donor Management	End-to-end new donor registration workflow	Registration page is accessible	1. Open registration page. 2. Fill in all required valid details. 3. Submit the form. 4. Log in as staff and view the donor list.	Name: Test Donor Email: donor3@test.com Phone: 9988776655 Blood Group: A+	Donor should be registered successfully and appear in the staff donor list	Not executed	Not Run
ST-DNTN-01	Blood Donation Management	End-to-end donation recording workflow	Donor is registered; staff is logged in	1. Staff opens donation recording page. 2. Selects donor and enters donation details. 3. Saves the donation. 4. Verifies updated inventory and donor history.	Blood Group: AB+ Quantity: 1 unit	Donation should be recorded, donor history updated, and inventory increased accordingly	Not executed	Not Run
ST-DNTN-02	Blood Donation Management	Donor views their own donation history	Donor has at least one recorded donation and is logged in	1. Log in as donor. 2. Navigate to donation history page.	N/A	Donor should see a chronological list of their past donations	Not executed	Not Run
ST-BG-01	Blood Group Management	Verify a selected blood group is consistently reflected across the workflow	A donation is being recorded	1. Select a blood group during donation entry. 2. Save. 3. Check the same blood group appears in inventory and donor history.	Blood Group: A- Quantity: 1 unit	Selected blood group should be consistently recorded and displayed across inventory and donor history	Not executed	Not Run
ST-BG-02	Blood Group Management	Verify all blood groups are listed in the inventory summary, including zero-stock groups	Inventory has recorded stock for multiple blood groups	1. Log in as staff. 2. Open the inventory summary page. 3. Check that all eight blood groups are listed.	N/A	Inventory summary should list all eight standard blood groups regardless of stock level	Not executed	Not Run
ST-INV-01	Blood Inventory Management	End-to-end staff review of blood inventory summary	Staff is logged in; inventory has data for multiple blood groups	1. Log in as staff. 2. Open inventory summary page. 3. Review unit counts per blood group.	N/A	Inventory summary should display accurate, up-to-date unit counts for each blood group	Not executed	Not Run
ST-INV-02	Blood Inventory Management	Verify administrator can record expiry information for a stored blood unit	Administrator is logged in; a blood unit exists	1. Log in as administrator. 2. Open the unit record. 3. Enter a valid expiry date. 4. Save.	Expiry Date: a valid future date	Expiry date should be saved and the unit should remain in available inventory until that date	Not executed	Not Run

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
ST-SRCH-01	Blood Availability Search	End-to-end availability search workflow for a requester	Requester is logged in	1. Log in as requester. 2. Open availability search page. 3. Select a blood group. 4. View results.	Blood Group: A+	Requester should see the current available unit count for the selected blood group	Not executed	Not Run
ST-SRCH-02	Blood Availability Search	Verify unauthenticated users cannot access the availability search page	User is not logged in	1. Attempt to open the availability search page URL directly without logging in.	N/A	User should be redirected to the login page rather than shown availability data	Not executed	Not Run
ST-REQ-01	Blood Request Management	End-to-end request submission and status tracking workflow	Requester is logged in	1. Submit a blood request. 2. Check status as Pending. 3. Have staff approve or reject it. 4. Recheck status.	Blood Group: B- Quantity: 1 unit	Request should move correctly from Pending to Approved/Rejected and be reflected to the requester	Not executed	Not Run
ST-REQ-02	Blood Request Management	Verify a request exceeding available stock is flagged for staff review	Available stock is less than the requested quantity	1. Submit a request for a quantity greater than available stock. 2. Staff reviews the request.	Blood Group: AB- Requested: 10 units (stock: 1 unit)	Staff should be able to see that the requested quantity exceeds available stock before approving	Not executed	Not Run
ST-ISS-01	Blood Issue and Distribution	End-to-end approve-and-issue workflow	A pending request exists; staff is logged in	1. Staff approves the request. 2. Staff issues blood against the approved request. 3. Verify inventory and request status.	Blood Group: A+ Quantity: 1 unit	Request should be marked Issued, inventory should be reduced, and a transaction record should exist	Not executed	Not Run
ST-ISS-02	Blood Issue and Distribution	Verify a rejected request cannot proceed to the issue stage	A rejected request exists	1. Attempt to issue blood against the rejected request.	Request Status: Rejected	System should prevent the issue action for a rejected request	Not executed	Not Run
ST-ADM-01	Administrative Management	End-to-end user account management workflow	Administrator is logged in	1. Create a new staff account. 2. Verify the account can log in. 3. Deactivate the account. 4. Verify login is now blocked.	Username: staff02 Password: Staff@123	Account should be created, usable, then correctly blocked after deactivation	Not executed	Not Run
ST-ADM-02	Administrative Management	Verify a non-administrator cannot access user account management pages	A donor or requester account is logged in	1. Attempt to open the administrative user management page directly via URL.	N/A	Access should be denied and the user redirected away from the administrative page	Not executed	Not Run